

MLOps

In-Class Assignment 5

Let's take a data science production system where we have the following steps, with probability distributions given for the length of time (in days) for the process:

- 1) Define the problem: uniform distribution on [3, 14].
- 2) Obtain data extract: exponential, mean of 10 days.
- 3) For model development, we have
 - a. Cleaning data, EDA, and feature engineering: normal, mean of 25 days, standard deviation of 5.
 - b. Develop model: normal, mean of 5 days, standard deviation of 1.
 - c. Get feedback, test, and retune model: 65 days, standard deviation of 2. (collecting on live data)
- 4) For model productionization, we have
 - a. Design architecture: exponential distribution, offset of 20 days, mean of 20 days after offset.
 - b. Development of data pipelines: normal distribution, mean of 30 days, standard deviation of 5.
 - c. Logging and monitoring productionization: uniform distribution on [10, 20]
- 5) Unit and integration testing: mean of 30 days, standard deviation of 6.

(3) and (4) can be done in parallel; everything else is in series. Run 1,000,000 simulations of this process.

What is the mean, standard deviation, and 95th quantile of the total time?

For what percent of times does development take longer than production?

How would you measure the greatest source of uncertainty in this project?

(time permitting) Suppose that, after integration testing, it fails 50% of the time, and you have to re-do this testing. Give the new mean, standard deviation, and 95th quantile for the time taken.